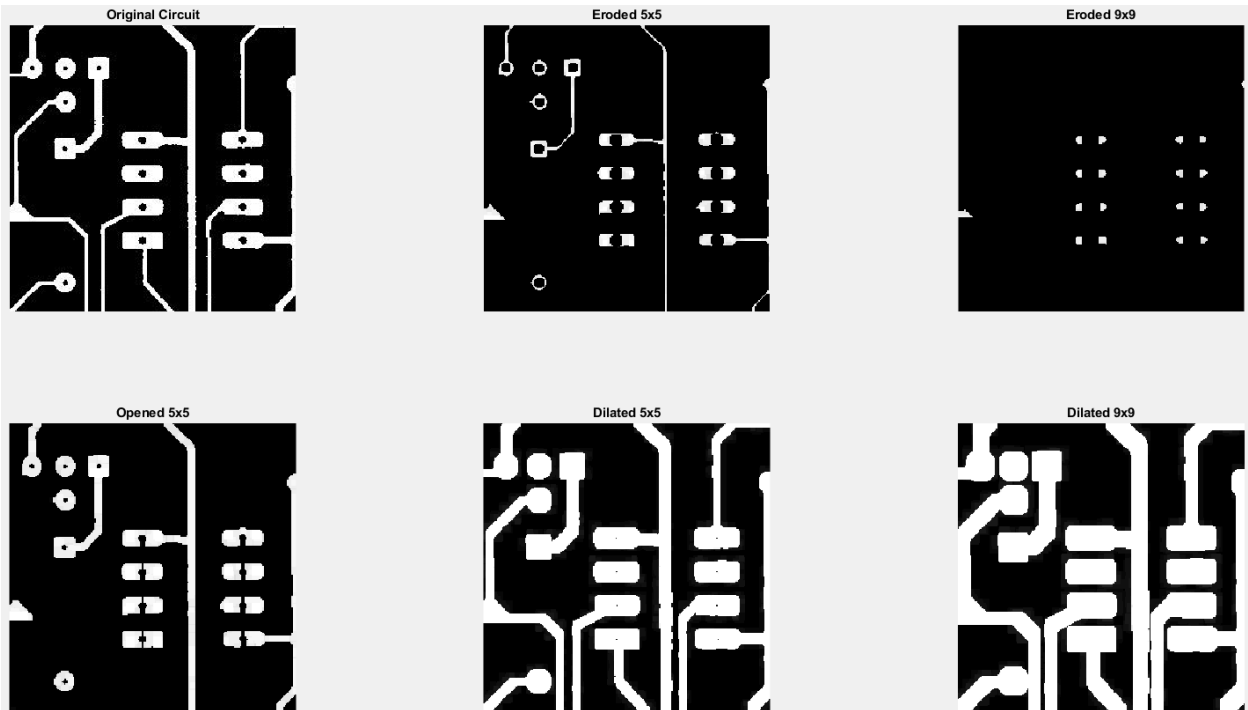


4TN4 Quiz 3

1. The following image shows erosion and dilation of the “Circuit Board” image from class. There is a clear loss of detail even with the 5x5 erosion, and the 9x9 erosion nearly gets rid of the whole image. With 5x5 dilation, there is obvious blow out but some smaller features are distinguishable. With 9x9 dilation this is not possible.



2. The above image also shows the image after performing opening, or in other words erosion followed by dilation using a 5x5 square structuring element. In theory, this should return the image back to its original condition, but as can be seen there is some loss of data and deletion of bounding details, leaving holes in the middle of the objects as well as loss of the thinner wires.
3. The image below shows a binarized version of the “leaf.tif” image from class. This was done using Otsu’s method, finding the between-class variance, and selecting threshold based on the maximum variance. There is also a histogram showing the optimal threshold overlaid. An interesting thing to note is the lack of true blacks in the original image.

